SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5108
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I A. Sai Joshitha (192424220) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Joshitha

Q1 RSA Encryption Analysis

Given,

$P = 11$, $q = 13$, $e = 7$, $M = 9$

Where:

- P, q = prime numbers
- e = public exponent
- M = original message
- n = modulus
- d = private exponent
- c = cipher text

(a) compute $\phi(n)$

Step 1 : calculate n.

Formula:

$$\boxed{n = P \times q}$$

Substitute the values:

$$n = 11 \times 13$$

$$\boxed{n = 143}$$

Step 2 : calculate Euler's totient function

Formula:

$$\boxed{\phi(n) = (P-1)(q-1)}$$

Substitute:

$$\phi(n) = (11+1)(13-1)$$

$$= 10 \times 12$$

$$\boxed{\phi(n) = 120}$$

Therefore,

$$\boxed{n = 143, \quad \phi(n) = 120}$$

(b) choosing $e = 7$ finding the private key d

The public exponent e must satisfy:

$$1 < e < \phi(n)$$

and

$$\boxed{\gcd(e, \phi(n)) = 1}$$

Here, $\gcd(7, 120) = 1$

Therefore, $\boxed{e = 7}$

step 1 : use the private-key formula

$$\boxed{ed = 1 \pmod{\phi(n)}}$$

Substitute:

$$7d = 1 \pmod{120}$$

This means $7d$ must leave a remainder of 1 when divided by 120

step 2 : Find d using Euclidean method

Divide 120 by 7 :

$$120 = 7 \times 17 + 1$$

Rearrange:

$$1 = 120 - 7 \times 17$$

Therefore,

$$7(-17) = 1 \pmod{120}$$

Since a positive value is required, add 120:

$$d = -17 + 120$$

$$\boxed{d = 103}$$

Step 3: By verifing

$$e \times d = 7 \times 103 = 721$$

Divide 721 by 120:

$$721 = 120 \times 6 + 1$$

The remainder is 1. Therefore,

$$7 \times 103 = 1 \pmod{120}$$

- public key : (7, 143)
- private key : (103, 143)

(c) Encrypt the message $M = 9$

step 1: write the encryption formula

$$\boxed{C = M^e \bmod n}$$

substitute:

$$c = 9^7 \bmod 143$$

step 2: calculate the powers of 9

$$9^2 = 81$$

$$9^3 = 9 \times 81 = 729$$

Divide 729 by 143:

$$729 = 143 \times 5 + 14$$

Therefore, $9^3 = 14 \pmod{143}$

Next,

$$9^4 = 14 \times 9 = 126 \pmod{143}$$

$$9^5 = 126 \times 9 = 1134$$

$$1134 = 143 \times 7 + 133$$

Hence, $9^5 = 133 \pmod{143}$

$$9^6 = 133 \times 9 = 1197$$

$$1197 = 143 \times 8 + 53$$

Therefore, $9^6 = 53 \pmod{143}$

Finally,

$$9^7 = 53 \times 9 = 477$$

$$477 = 143 \times 3 + 48$$

Thus,

$$\boxed{C = 48}$$

The cipher text is 48

(d) Decrypt the ciphertext and verify the result

use the cipher text calculated in part (c),

$C = 48$

Step 1 : Write the decryption formula

$$\boxed{M = c^d \bmod n}$$

Substitute:

$$M = 48^{103} \bmod 143$$

Step 2 : calculate using repeated squaring.

$$48^2 = 2304$$

$$2304 = 143 \times 16 + 16$$

Therefore,

$$48^2 = 16 \pmod{143}$$

$$48^4 = 16^2 = 256$$

$$256 = 143 + 113$$

$$\Rightarrow 48^4 = 113 \pmod{143}$$

$$\Rightarrow 48^8 = 113^2 = 12769$$

$$12769 = 143 \times 89 + 42$$

$$48^8 = 42 \pmod{143}$$

$$\Rightarrow 48^{16} = 42^2 = 1764$$

$$1764 = 143 \times 12 + 48$$

$$48^{16} = 48 \pmod{143}$$

$$\Rightarrow 48^{32} = 48^2 = 16 \pmod{143}$$

$$48^{64} = 16^2 = 113 \pmod{143}$$

Step 3: Express the exponent 103 as a sum of powers of 2

$$103 = 64 + 32 + 4 + 2 + 1$$

Therefore,

$$48^{103} = 48^{64} \times 48^{32} \times 48^4 \times 48^2 \times 48$$

Taking modulo 143:

$$M = 113 \times 16 \times 113 \times 16 \times 48 \pmod 1$$

by Simplifying,

$$113 \times 16 = 1808$$

$$1808 = 143 \times 12 + 92$$

So,

$$M = 92 \times 113 \times 16 \times 48 \pmod{143}$$

$$92 \times 113 = 10396$$

$$10396 = 143 \times 72 + 100$$

Thus,

$$M = 100 \times 16 \times 48 \pmod{143}$$

$$\Rightarrow 100 \times 16 = 1600$$

$$1600 = 143 \times 11 + 27$$

Thus,

$$M = 27 \times 48 \pmod{143}$$

$$27 \times 48 = 1296$$

$$1296 = 143 \times 9 + 9$$

Therefore,

$$\boxed{M = 9}$$

original message = 9

Decrypted message = 9

Both are equal, so RSA encryption and decryption are verified

(e) Analyze how changing e affects security

The encryption formula is:

$$\boxed{C = M^e \bmod n}$$

changing e changes the encryption calculation and the corresponding private key d.

1. Small e : Encryption can be faster, but improper padding or careless use can expose messages to certain attacks.

2. Large e : Encryption may require more computation, depending on the exponent's binary representation.

3. Valid e : It must satisfy $\gcd(e, \phi(n)) = 1$, so that the private exponent exists.

4. Invalid e : If the greatest common divisor is not 1, a valid modular inverse d does not exist.

5. overall security : security mainly depends on sufficiently large RSA keys, secure padding, proper key management and implementation.

conclusion :

The exponent $e = 7$ is valid for this mathematical example. However, $n = 143$ is too small for practical security. Real Applications need appropriately sized RSA keys and secure padding.

Q4. Diffie - Hellman Key Exchange Analysis

Given,

$p = 23$, $g = 5$

$a = 6$, $b = 15$

Where:

- P = prime number
- g = generator
- a = user A's private key
- b = user B's private key
- A = user A's public key
- B = user B's public key
- K = shared secret key

(a) Given Parameters

The public parameters are:

$$\boxed{P = 23,\ g = 5}$$

The prime number is 23 and the generator is 5. Both users know these values.

(b) user A's private key

Given:

$$\boxed{a = 6}$$

user A keeps the private key secret and does not send it over the network

(c) user B's private key

Given:

$$\boxed{b = 15}$$

user B also keeps the private key secret

(d) Compute the public keys for A and B

User A's public key

Step 1 : write the formula

$$\boxed{A = g^a \bmod P}$$

Step 2 : Substitute the values

$$A = 5^6 \bmod 23$$

Step 3 : Calculate the powers

$$5^2 = 25$$

$$25 = 23 + 2$$

Therefore,

$$5^2 = 2 \pmod{23}$$

$$\Rightarrow 5^4 = 2^2 = 4 \pmod{23}$$

Now,

$$5^6 = 5^4 \times 5^2$$

$$5^6 = 4 \times 2 = 8 \pmod{23}$$

Therefore,

$$\boxed{A = 8}$$

User B's public key

Step 1 : Write the formula

$$\boxed{B = g^b \bmod p}$$

Step 2 : substitute the values

$$B = 5^{15} \bmod 23$$

Step 3 : calculate using repeated squaring

$$5^2 = 2 \pmod{23}$$

$$5^4 = 2^2 = 4 \pmod{23}$$

$$5^8 = 4^2 = 16 \pmod{23}$$

Since,

$$15 = 8+4+2+1$$

we get,

$$5^{15} = 5^8 \times 5^4 \times 5^2 \times 5$$

Taking modulo 23:

$$5^{15} = 16 \times 4 \times 2 \times 5 \pmod{23}$$

By calculating:

$$16 \times 4 = 64$$

$$64 = 23 \times 2 + 18$$

Remainder = 18

$$18 \times 2 = 36$$

$$36 = 23 + 13$$

Remainder = 13

$$13 \times 5 = 65$$

$$65 = 23 \times 2 + 10$$

Therefore,

$$\boxed{B = 19}$$

Finally,

$$\boxed{A = 8,\ B = 19.}$$

These public keys are exchanged between the users

(e) compute the shared secret key

After exchanging public keys, both users calculate the secret independently.

user A calculate the shared secret

Formula:

$$\boxed{K_A = B^a \bmod P}$$

Substitute:

$$K_A = 19^6 \bmod 23$$

Calculate:

$$19^2 = 361$$

$$361 = 23 \times 15 + 16$$

Therefore,

$$19^2 = 16 \pmod{23}$$

$$\Rightarrow 19^4 = 16^2 = 256$$

$$256 = 23 \times 11 + 3$$

Therefore,

$$19^4 = 3 \pmod{23}$$

$$\Rightarrow 19^6 = 19^4 \times 19^2$$

$$K_A = 3 \times 16 = 48 \pmod{23}$$

$$48 = 23 \times 2 + 2$$

Therefore,

$$\boxed{K_A = 2}$$

user B calculates the shared secret

Formula:

$$\boxed{K_B = A^b \bmod p}$$

Substitute:

$$K_B = 8^{15} \bmod 23$$

Calculate:

$$8^2 = 64 = 18 \pmod{23}$$

$$8^4 = 18^2 = 324$$

$$324 = 23 \times 14 + 2$$

Therefore,

$$8^4 = 2 \pmod{23}$$

$$\Rightarrow 8^8 = 2^2 = 4 \pmod{23}$$

Since,

$$15 = 8+4+2+1$$

We get,

$$8^{15} = 8^8 \times 8^4 \times 8^2 \times 8$$

Taking modulo 23:

$$K_B = 4 \times 2 \times 18 \times 8 \pmod{23}$$

By calculating,

$$4 \times 2 = 8$$

$$8 \times 18 = 144$$

$$144 = 23 \times 6 + 6$$

Remainder = 6.

$$6 \times 8 = 48$$

$$48 = 23 \times 2 + 2$$

Therefore, $\boxed{K_B = 2}$

By verifying,

$$K_A = K_B$$

$$\boxed{K = 2}$$

Conclusion:

Both users independently generate the same shared Secret Key, $K = 2$, without sending their private keys over the network.

(f) Demonstrate how a Man-in-the-Middle (MITM) attacker could after the exchange

A man-in-the-middle attacker intercepts and replaces public keys when the exchange is not authenticated.

Step 1: Normal exchange

- user A sends $A = 8$.
- user B sends $B = 19$.

Step 2: The attacker replace the public keys

Suppose the attacker chooses fake public values:

$$X = 10, \quad Y = 7$$

The attacker sends $X = 10$ to user B instead of $A = 8$, and sends $Y = 7$ to user A instead of $B = 19$

Step 3: User A calculates A secret with the attacker

$$K_{A,attacker} = 7^6 \bmod 26$$

Calculate:

$$7^2 = 49 = 3 \pmod{23}$$

$$7^4 = 3^2 = 9 \pmod{23}$$

$$7^6 = 7^4 \times 7^2$$

$$7^6 = 9 \times 3 = 27 = 4 \pmod{23}$$

Thus,

$$K_{A,\text{attacker}} = 4$$

Step 4 : user B calculates a secret with the attacker

$$K_{B,\text{attacker}} = 10^{15} \bmod 23$$

Calculate:

$$10^2 = 100 = 8 \pmod{23}$$

$$10^4 = 8^2 = 64 = 18 \pmod{23}$$

$$10^8 = 18^2 = 324 = 2 \pmod{23}$$

Since, $15 = 8+4+2+1$,

$$10^{15} = 2 \times 18 \times 8 \times 10 \pmod{23}$$

$$2 \times 18 = 36 = 13 \pmod{23}$$

$$13 \times 8 = 104 = 12 \pmod{23}$$

$$12 \times 10 = 120 = 5 \pmod{23}$$

Thus,

$$K_{B,\text{attacker}} = 5$$

Step 5 : Explain the attack

The attacker can establish one secret with user A and another with user B. If the attacker can decrypt and re-encrypt the

traffic, they may read or modify messages without either user noticing.

Conclusion:

Basic Diffie-Hellman does not authenticate the Public keys. Therefore, an attacker can replace them and establish separate secrets with the two users.

(g) Suggest a secure method to prevent the attack

These methods helps to prevent MITM attacks:

1. Digital certificates: verify the identities of the communicating users or servers through a trusted certificate Authority.

2. Digital signature: sign the exchanged public keys or handshake messages so that tampering can be detected.

3. Authenticated Diffie-Hellman: use an authenticated key-exchange protocol instead of unauthenticated Diffie-Hellman.

4. HTTPS/TLS: use properly configured TLS with certificate verification to authenicate the server and protect data in transit

5. public-key validation: Reject unverified or unexpected public keys

Conclusion :

Authenticated Diffie - Hellman combined with TLS and certificate Validation is a suitable practical solution. it helps to ensure that public keys belong to the intendend parties and prevents an attacker from silently replacing them.